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(54) **DEPLOYING A DOWNHOLE SAFETY VALVE WITH AN ARTIFICIAL LIFT SYSTEM**

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(58) **Field of Classification Search**

CPC E21B 43/128; E21B 31/00; E21B 33/12; E21B 34/06; E21B 2200/05

See application file for complete search history.

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Primary Examiner — Steven A MacDonald

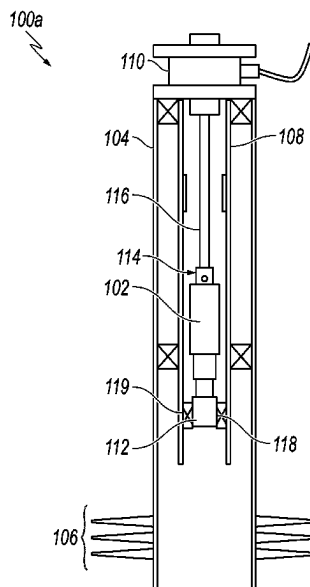
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ABSTRACT

A fluid end is coupled to and configured to be driven by an electric motor. A shear interconnect is at an uphole end of the electric submersible pump. The shear interconnect is configured to shear a cable line between the electric submersible pump and a topside facility. The shear interconnect is configured to shear the cable at the electric submersible pump. A safety valve is arranged to cease flow within a wellbore, in which the electric submersible pump is installed, when the safety valve is in a closed position.

13 Claims, 10 Drawing Sheets



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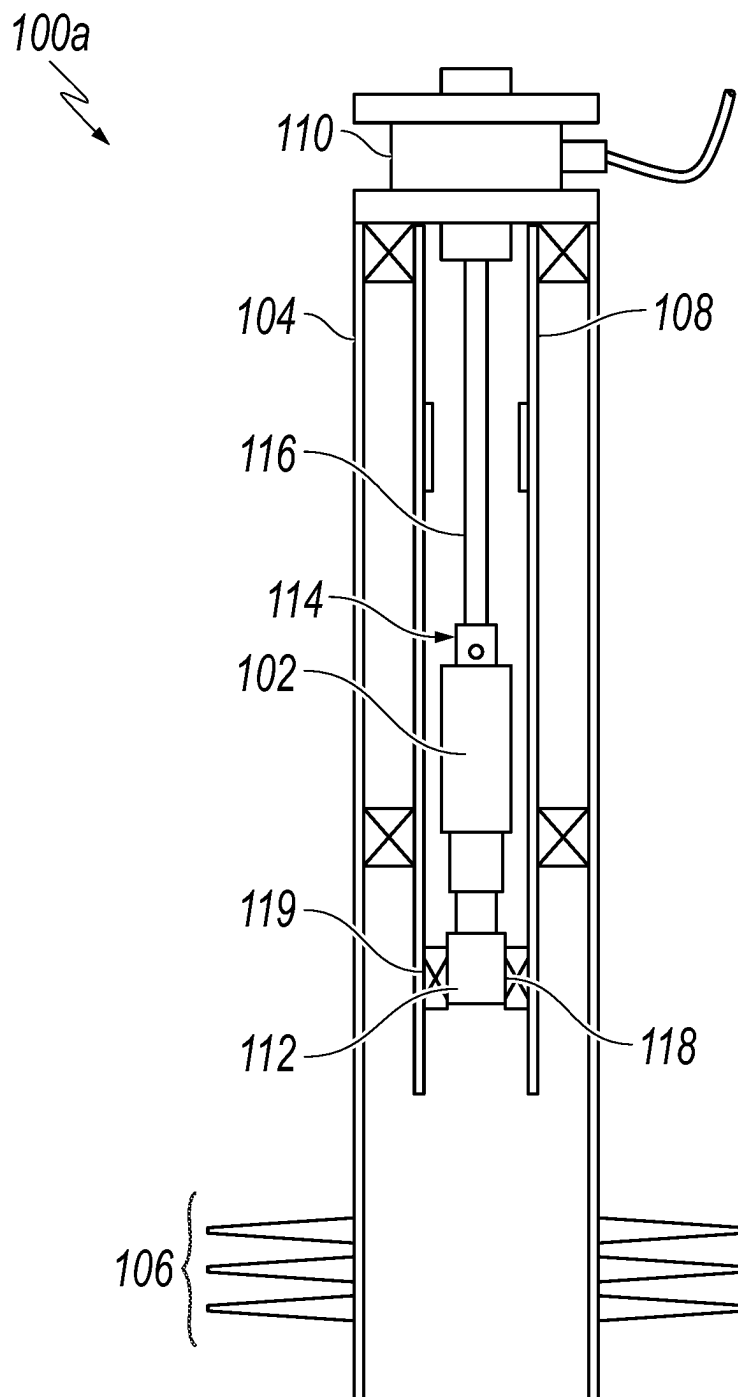


FIG. 1A

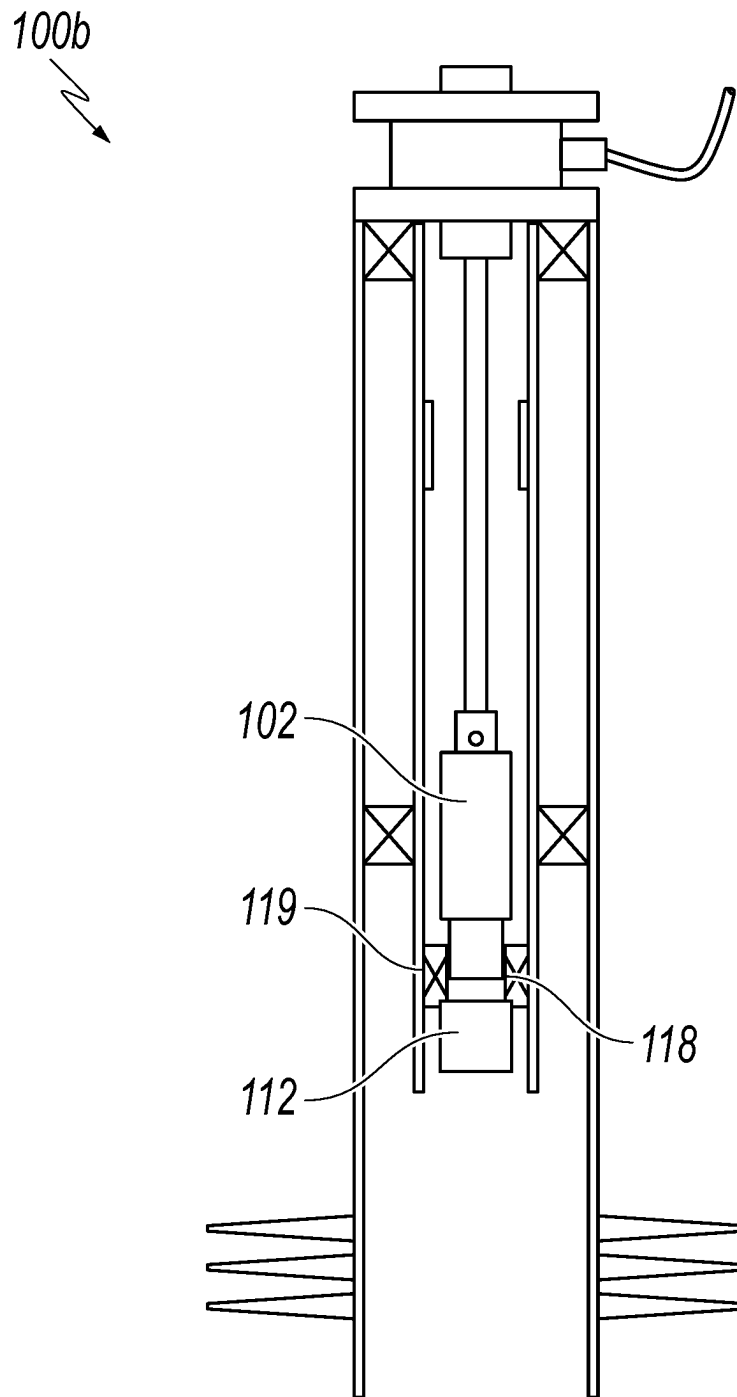


FIG. 1B

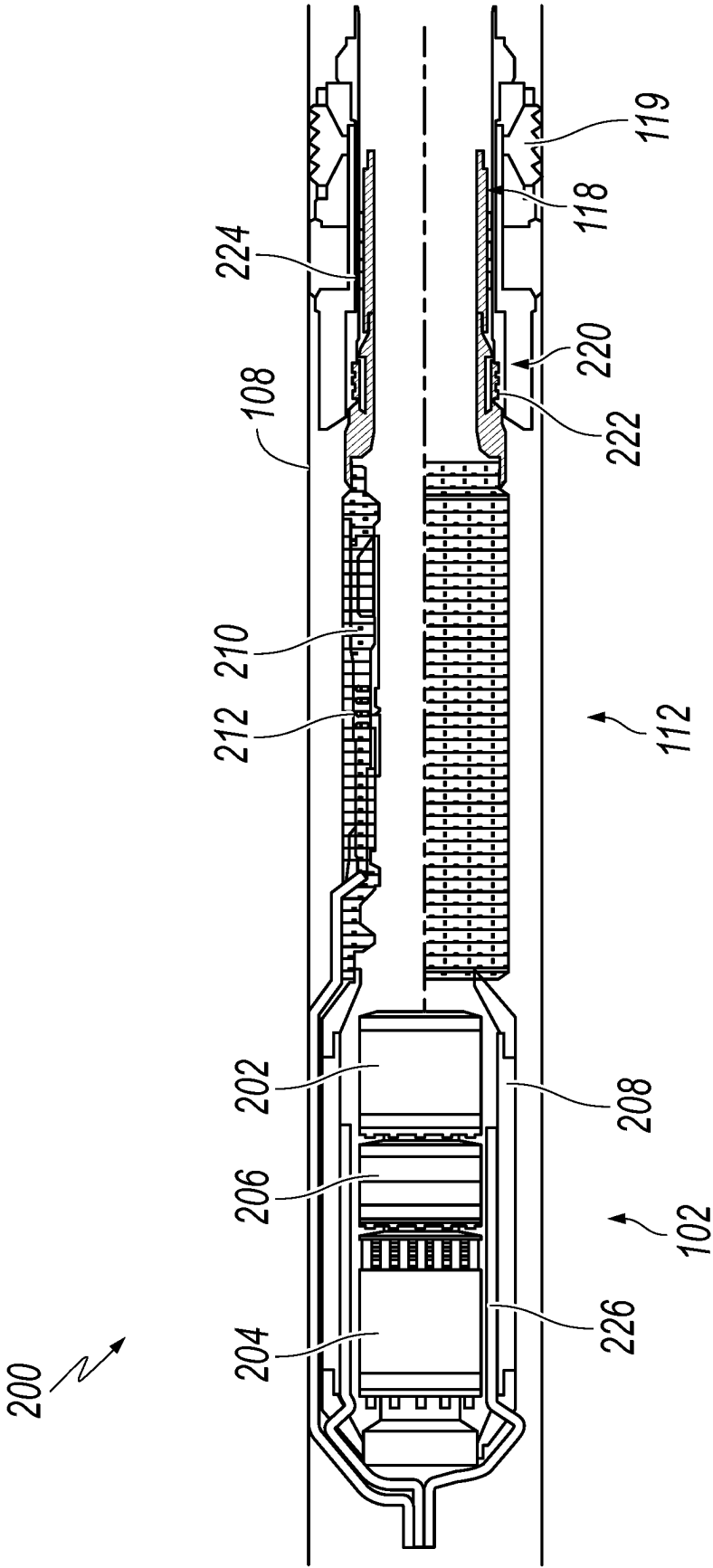


FIG. 2A

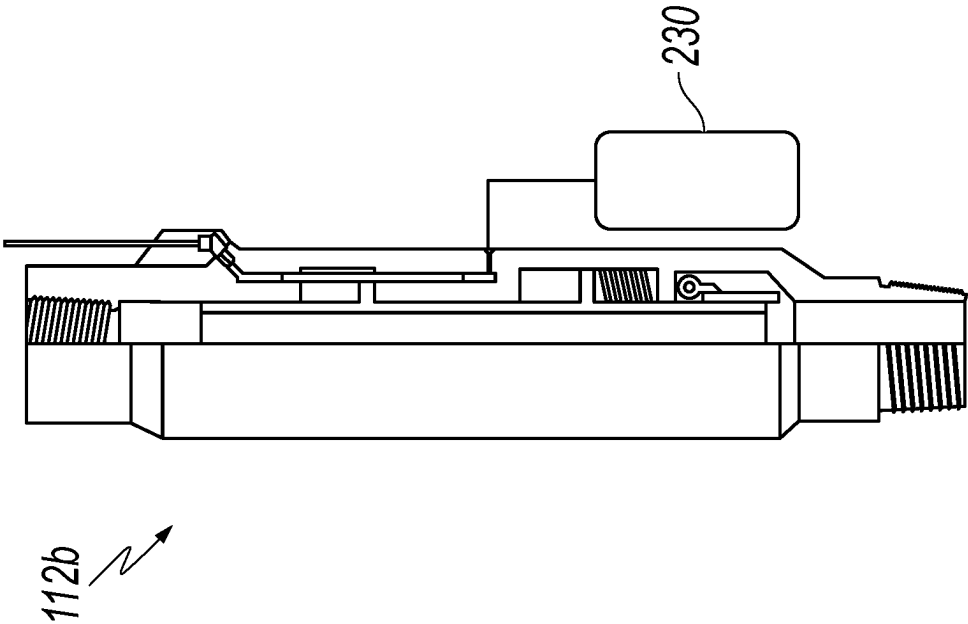


FIG. 2C

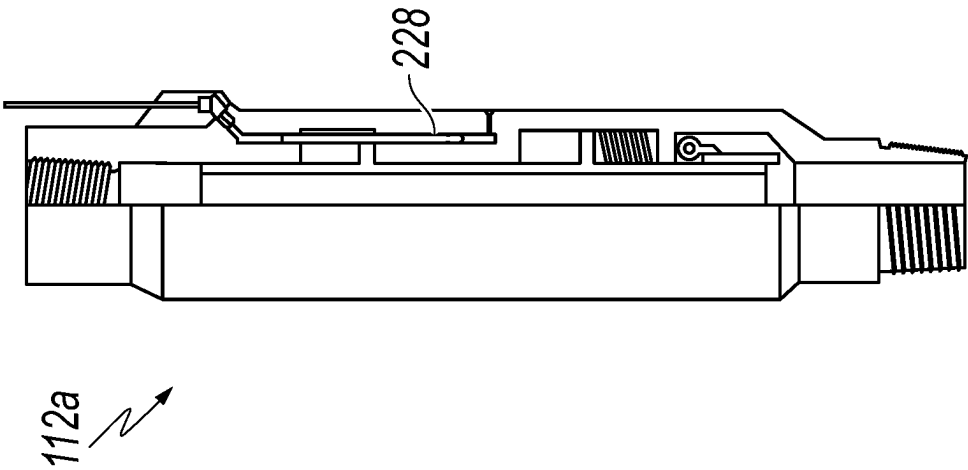


FIG. 2B

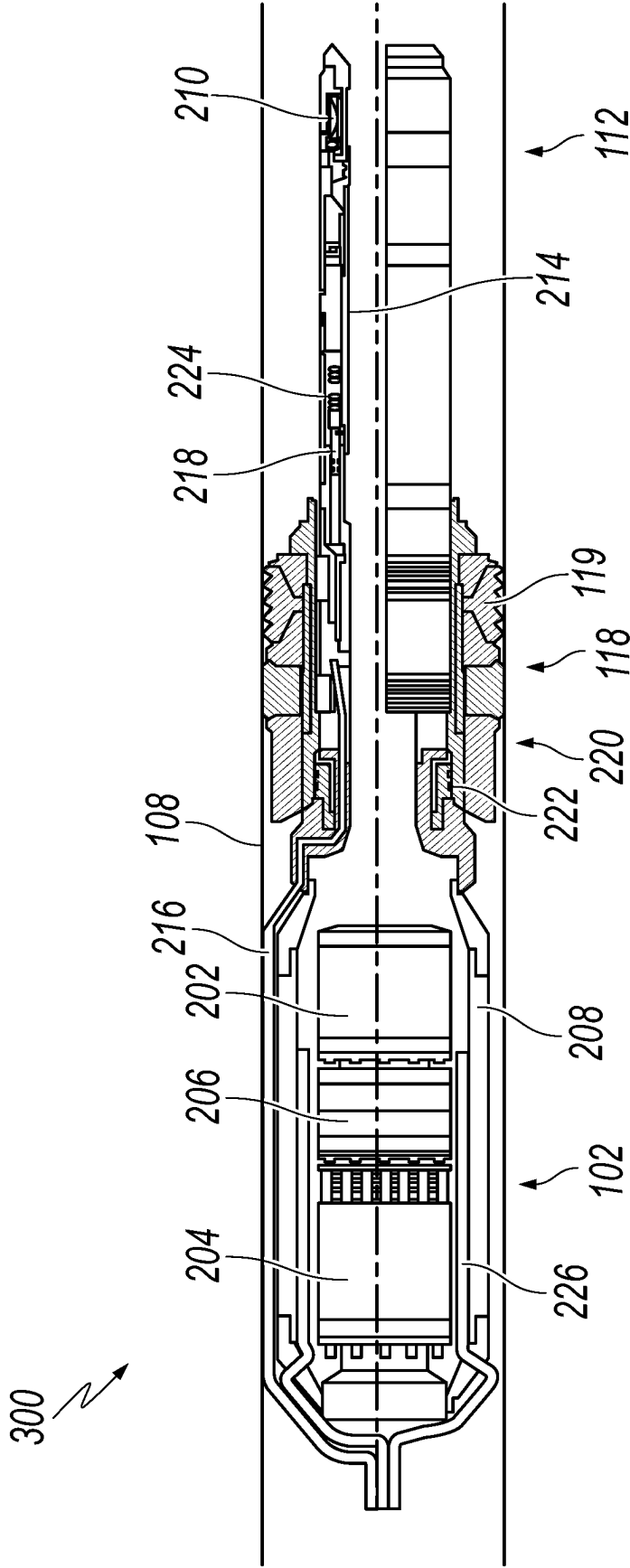


FIG. 3A

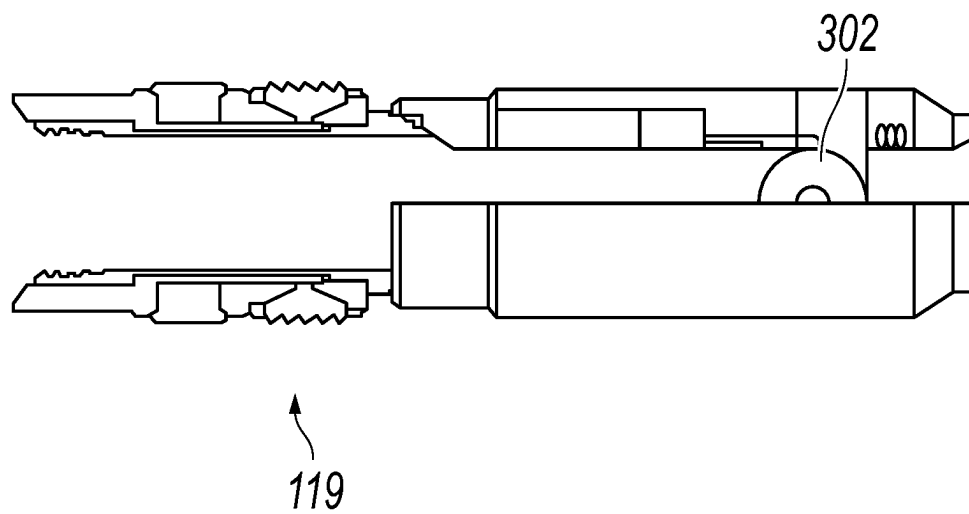


FIG. 3B

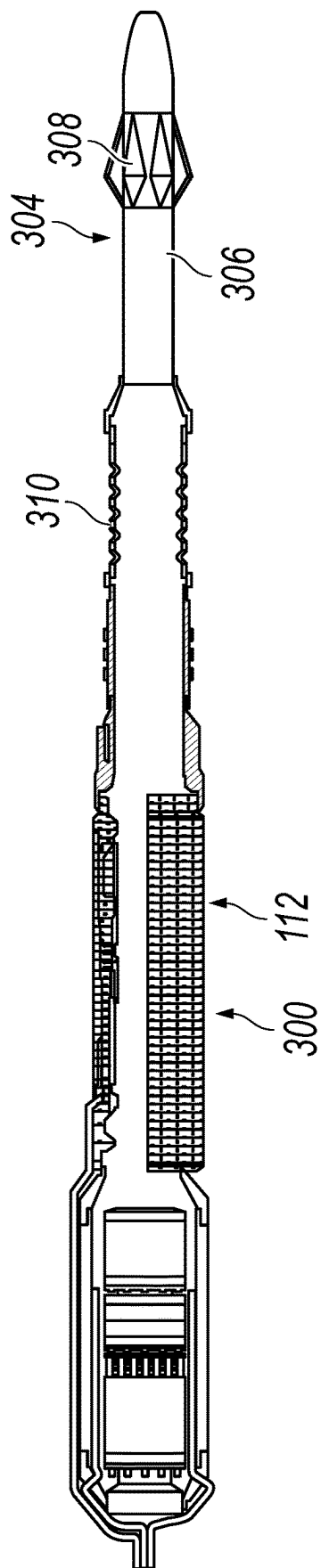


FIG. 3C

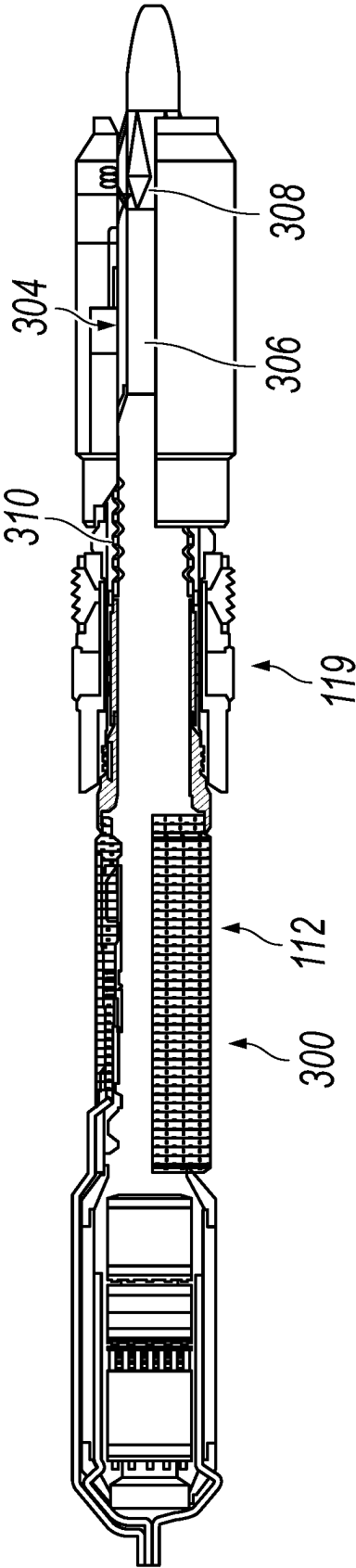


FIG. 3D

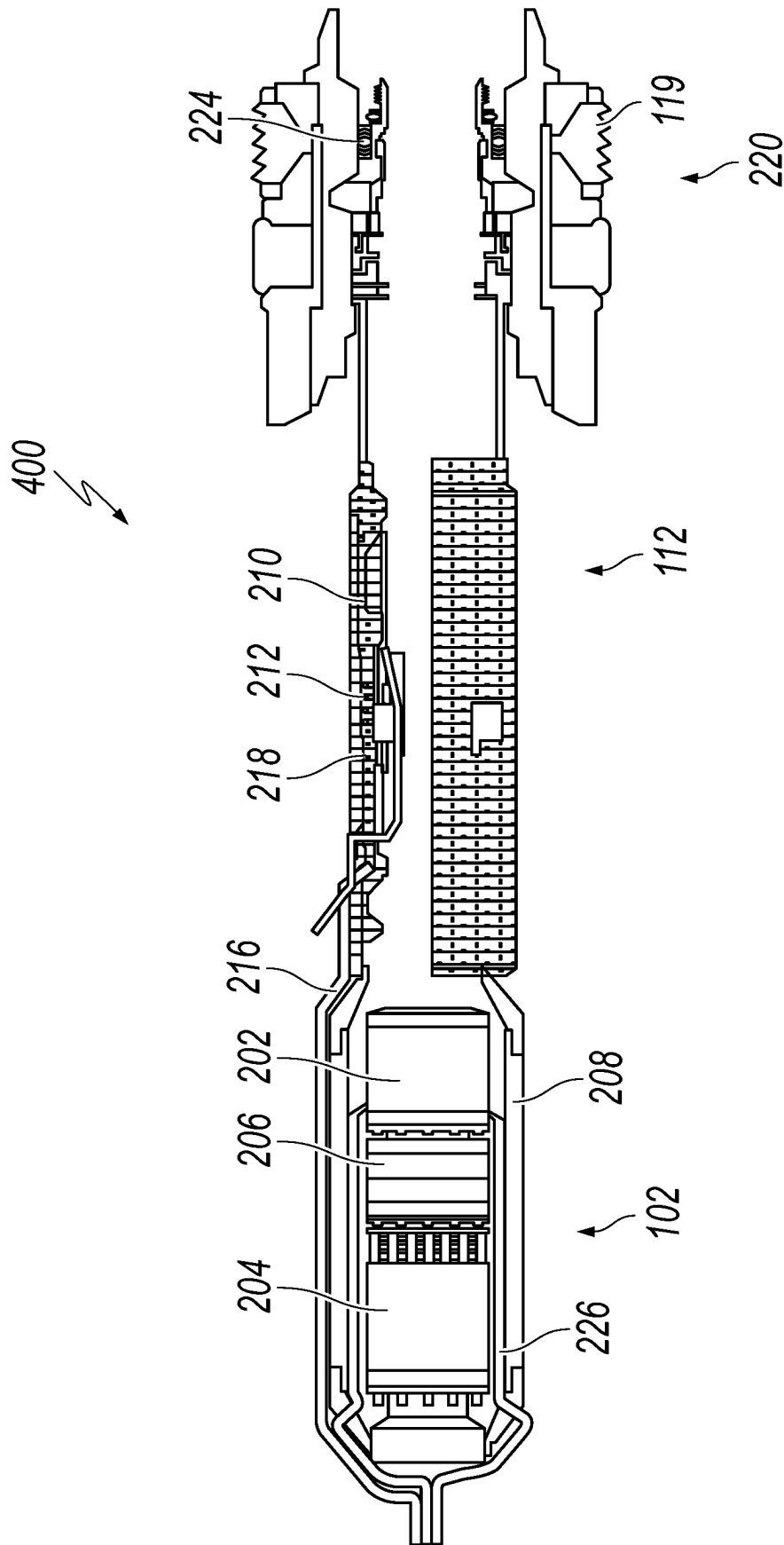


FIG. 4

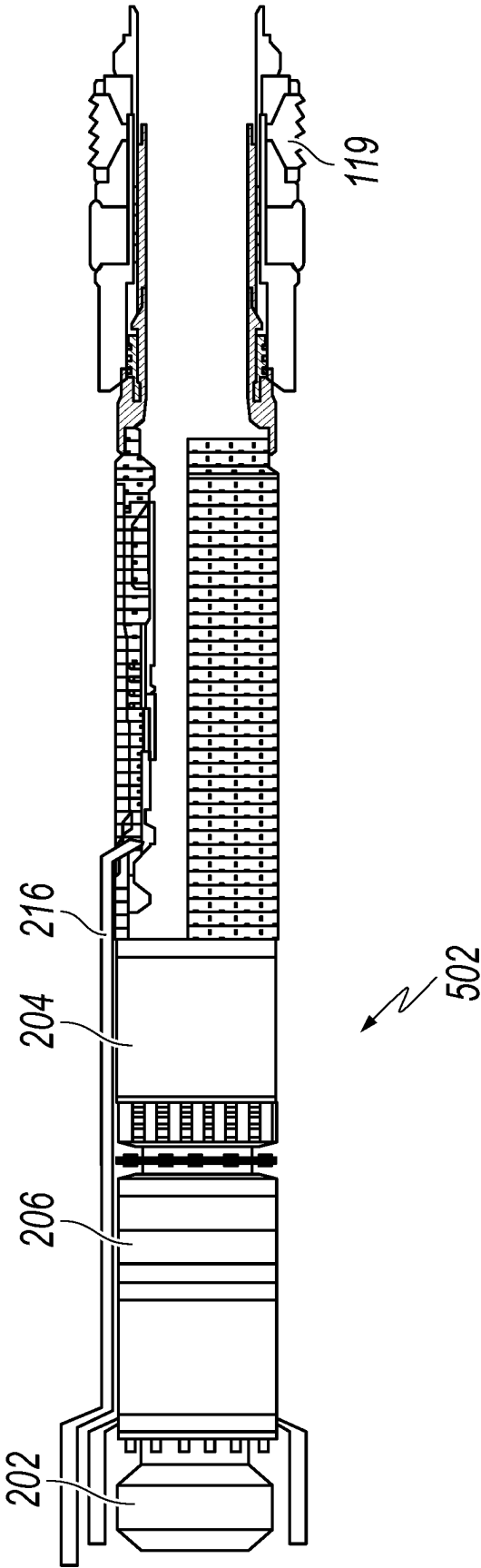


FIG. 5

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DEPLOYING A DOWNHOLE SAFETY VALVE WITH AN ARTIFICIAL LIFT SYSTEM

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

This application claims the benefit of priority U.S. application Ser. No. 17/661,692, filed May 2, 2022, which claims the benefit of priority U.S. Provisional Application No. 63/268,960, filed Mar. 7, 2022, the contents of which are incorporated by reference herein.

TECHNICAL FIELD

This disclosure relates to downhole safety valves and artificial lift system.

BACKGROUND

Most wells behave characteristically different over time due to geophysical, physical, and chemical changes in the subterranean reservoir that feeds the well. For example, it is common for well production to decline. This decline in production can occur due to declining pressures in the reservoir, and can eventually reach a point where there is not enough pressure in the reservoir to economically realize production through the well to the surface. Downhole pumps and/or compressors can be deployed into the well to increase production. Additionally or alternatively, a topside compressor and/or pump are sometimes used to extend the life of the well by decreasing pressure at the top of the well.

SUMMARY

This disclosure relates to deploying a downhole safety valve with an artificial lift system.

An example implementation of the subject matter described within this disclosure is an electric submersible artificial lift system with the following features. A fluid end is coupled to and configured to be driven by an electric motor. A shear interconnect is at an uphole end of the electric submersible artificial lift system. The shear interconnect is configured to shear a cable extending between the electric submersible artificial lift system and a topside facility. The shear interconnect is configured to shear the cable at the electric submersible artificial lift system. A safety valve is arranged to cease flow within a wellbore, in which the electric submersible artificial lift system is installed, when the safety valve is in a closed position.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The electric submersible artificial lift system is configured to be stabbed into and seal with a downhole receptacle.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. A shroud encapsulates the electric motor and the fluid end. The shroud is configured to be stabbed into the downhole receptacle.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The safety valve includes a flapper valve biased towards a closed position.

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Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The electric motor and the fluid end are coupled together by a magnetic coupling.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The safety valve is at an uphole end of the fluid end.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The safety valve is downhole of the fluid end.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The safety valve is integrated into the electric submersible artificial lift system.

Aspects of the example electric submersible artificial lift system, which can be combined with the electric submersible artificial lift system alone or with other aspects, include the following. The electric motor is uphole of the fluid end.

An example implementation of the subject matter described within this disclosure is a method with the following features. An electric submersible artificial lift system is received by a packer that includes a receptacle configured to receive the electric submersible artificial lift system. A safety valve is received by the packer.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. The packer includes a latch. The method further includes securing, by the latch, the electric submersible artificial lift system to the packer.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. The electric submersible artificial lift system is released by the latch.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. Releasing the electric submersible artificial lift system includes receiving a fishing tool by the electric submersible artificial lift system.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. The electric submersible artificial lift system is over-pulled or jarred by the fishing tool. The latch released the electric submersible artificial lift system responsive to the over-pull or jar.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. The safety valve is left within a wellbore with the packer.

Aspects of the example method, which can be combined with the example method alone or with other aspects, include the following. A cable at an uphole end of the electric submersible artificial lift system is sheared.

An example implementation of the subject matter described within this disclosure is a wellbore system with the following features. A cable extends into a wellbore from a topside facility. The cable includes electrical lines, hydraulic lines, and a support structure configured to support tooling at a downhole end of the cable. An electric submersible artificial lift system is at the downhole end of the cable. The electric submersible artificial lift system includes the following features. A fluid end is coupled to and configured to be driven by an electric motor. A shear interconnect is at

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an uphole end of the electric submersible artificial lift system. The shear interconnect is configured to shear a cable between the electric submersible artificial lift system and the topside facility. The shear interconnect configured to shear the cable at the electric submersible artificial lift system. A packer defines a receptacle configured to receive the electric submersible artificial lift system. A safety valve is arranged to cease flow within a wellbore, in which the electric submersible artificial lift system is installed, when the safety valve is in a closed position.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The hydraulic line is configured to actuate the safety valve and inject chemicals into a production stream.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The hydraulic line is configured to deliver lubrication to the electric submersible artificial lift system and actuate the safety valve.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The safety valve is integrated with the electric submersible artificial lift system.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. A mechanical shifting tool is configured to actuate an isolation barrier within the receptacle. The isolation barrier is biased towards a closed position. The mechanical shifting tool is configured to move the isolation barrier to an open position once the electric submersible artificial lift system and safety valve are received by the receptacle.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The safety valve is integrated into the packer.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The packer includes a latch configured to secure the electric submersible artificial lift system.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. A balance line is fluidically arranged to transfer pressure from a portion of the wellbore downhole of the valve to the safety valve. Pressure from the balance line reduces a force needed to actuate the safety valve.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. A pressurized canister provides pressure to the safety valve. The pressure from the pressurized canister reduces a force needed to actuate the safety valve.

Aspects of the example wellbore system, which can be combined with the example wellbore system alone or with other aspects, include the following. The safety valve is an electric safety valve coupled to a power system of the electric submersible artificial lift system. The electric safety valve is configured to operate responsive to power provided to the electric submersible artificial lift system.

The details of one or more implementations of the subject matter described within this disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the subject matter

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described herein will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

FIG. 1A is a side cross-sectional diagram of an example downhole artificial lift arrangement.

FIG. 1B is a side cross-sectional diagram of an example downhole artificial lift arrangement.

FIG. 2A is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement.

FIGS. 2B and 2C are quarter cross-sectional views of example safety valves.

FIG. 3A is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement.

FIGS. 3B-3D are side cross-sectional diagrams of example packer and receptacle arrangements that can be used with aspects of this disclosure.

FIG. 4 is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement.

FIG. 5 is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

This disclosure describes an artificial lift arrangement that allows for easy retrieval and repair of artificial lift systems, such as electric submersible pumps, and downhole safety valves. The implementations described herein include an electric submersible pump, a downhole safety valve, and a receptacle configured to receive and retain the electric submersible pump. The safety valve is integrated with the electric submersible pump or the receptacle. The electric submersible pump is coupled to a wellhead or other topside equipment by a cable that includes electrical and/or hydraulic lines. The electric and/or hydraulic lines power and control the electric submersible pump and/or the safety valve.

FIG. 1A is a side cross-sectional diagram of an example downhole artificial lift arrangement 100a. The artificial lift arrangement 100a includes an electric submersible pump 102 within a wellbore 104. During operation, the electric submersible pump 102 assists in flowing production fluid from a production zone 106, up through production tubing 108, to a topside facility, such as a wellhead 110. In some implementations, the topside facility includes a subsea wellhead. While this disclosure primarily describes implementations using electric submersible pumps, other downhole artificial lift systems, such as electric submersible compressors, top-driven pumps or compressors, plunger pumps and compressors, and gerotor pumps can be used without departing from this disclosure. Other mechanical lift devices, including positive displacement and centrifugal fluid movers, can be used without departing from this disclosure.

The electric submersible pump 102 includes an integrated safety valve 112 configured to cease fluid flow through the wellbore 104 when the safety valve 112 is in the closed position. While the illustrated implementation shows the safety valve to be at a downhole end of the electric submersible pump 102, other arrangements are possible without departing from this disclosure. More details on various implementations of the safety valve 112 are described throughout this disclosure.

In some instances, such as during deployment, the cable supports the electric submersible pump. The cable includes

electric and/or hydraulic lines **116** as well as any structural components to ensure the cable supports the weight of the electric submersible pump **102**. The cable includes a smooth outer surface such that the cable can be fed through a lubricator or similar structure. At an uphole end of the electric submersible pump **102** is a shear interconnect **114**. The shear interconnect **114** allows for a cable with electric and/or hydraulic lines **116** and structural components, extending between the wellhead **110** and the electric submersible pump **102**, to be sheared at an uphole end of the electric submersible pump **102**. That is, the electric and/or hydraulic lines **116** are sheared at the shear interconnect **114** such that the electric and/or hydraulic lines **116** can be removed completely or nearly completely from the wellbore while the electric submersible pump **102** remains within the wellbore **104**. The shear interconnect **114** initiates a shearing action when the electric and/or hydraulic lines **116** experience an over-pull scenario. That is, the shearing action occurs when tension within the electric and/or hydraulic lines exceeds a specified threshold. Typically, the tension is increased by pulling an uphole end of the electric and/or hydraulic lines **116** in an uphole direction at the wellhead **110**. While primarily described within this disclosure as decoupling by a shearing action, other disconnection mechanisms can be used to disconnect the cable from the electric submersible pump without departing from this disclosure. For example, a hydraulically controlled cable release system can be used. Such an implementation includes a latching mechanism to secure and retain the cable to the electric submersible pump, and a signal (e.g. hydraulic, pneumatic, or electric) is used to actuate the latch to a secure position or a release position.

The electric submersible pump **102** is stabbed into a downhole receptacle **118**. The downhole receptacle **118** can include a polished bore receptacle, a packer configured to receive the electric submersible pump, or any other receptacle that is appropriate for the operations described herein. The electric submersible pump **102** and the downhole receptacle **118** seal against one another when the electric submersible pump is fully received by the downhole receptacle **118**. That is, fluid flows primarily through the downhole receptacle **118** and electric submersible pump **102** with little-to-no leakage past the seals (shown in later figures). In some implementations, the downhole receptacle **118** includes a latch configured to secure and retain the electric submersible pump **102** within the downhole receptacle **118**. Such latches are described in greater detail later within this disclosure. In implementations where the safety valve **112** is integrated into the downhole end of the electric submersible pump, the safety valve **112** is the portion of the electric submersible pump stabbed into and received by the downhole receptacle **118**.

FIG. 1B is a side cross-sectional diagram of an example downhole artificial lift arrangement **100b**. The artificial lift arrangement **100b** is substantially similar to the artificial lift arrangement **100a** with the exception of any differences described herein. A safety valve **112** is integrated into the downhole receptacle **118**. The safety valve **112** is biased in the closed position when the electric submersible pump **102** is out of the downhole receptacle **118**. In this implementation, the electric submersible pump is directly stabbed into the downhole receptacle **118**.

FIG. 2A is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement **200**. The electrical submersible pump and safety valve arrangement **200** share similar components and arrangements to the artificial lift arrangement **100a**, and, in some

instances, can be used within the artificial lift arrangement **100a**. The electrical submersible pump and safety valve arrangement **200** includes the electric submersible pump **102**. The electric submersible pump **102** includes an electric motor **202** coupled to a fluid end **204** by a coupling **206**. The fluid end **204** is configured to be driven by the electric motor **202**. In some implementations, the coupling **206** includes a magnetic coupling or a direct-drive coupling.

The electric submersible pump **102** includes a shroud **208** encapsulating the electric motor **202**, the coupling **206**, and the fluid end **204**. At a downhole end of the shroud **208** is the safety valve **112**. The safety valve **112** includes a flapper **210** biased to close in the uphole direction, for example, by a spring **212**. The flapper **210** is actuated towards the open position by a sleeve **214** along an inner surface of a flow passage defined by the safety valve **112**. The sleeve **214** can be moved in a downhole direction to open the flapper in a variety of ways.

For example, a hydraulic line **216** from a topside facility that can apply pressure to a piston **218** to overcome the bias and open the flapper by moving the sleeve. In some implementations, the hydraulic line **216** used to operate the safety valve **112** is also used for chemical injection. For example, when pressure is provided to the hydraulic line **216** at a first specified pressure, the safety valve is opened in response to the provided pressure. In some implementations, when pressure is provided a second specified pressure, greater than the first specified pressure, fluid is injected into the production fluid while holding the valve open. In some implementations, such an arrangement is used to inject scale inhibitor, wax inhibitor, scale inhibitor, or hydrate inhibitor. Other chemicals can be injected without departing from this disclosure. Alternatively or in addition, in some implementations, the hydraulic line **216** is used to provide lubricant to the electric submersible pump and to provide pressure to actuate the safety valve **112**.

In some implementations, the safety valve **112** is mechanically operated, for example, by a linkage that moves the flapper to the open position when the electric submersible pump **102** is stabbed into the downhole receptacle **118**. In some implementations, the safety valve **112** is an electric safety valve coupled to a power system (e.g. electrical lines **226**, windings of the electric motor, or onboard control circuitry) of the electric submersible pump **102**. In such implementations, the electric safety valve is configured to operate responsive to power provided to the electric submersible pump **102**. For example, when the electric submersible pump is running, the electric safety valve opens in response.

FIGS. 2B and 2C are quarter cross-sectional views of example safety valves **112a** and **112b**. The safety valves **112a** and **112b** are substantially similar to safety valve **112**, with the exception of any difference described herein, and can be used in lieu of safety valve **112** in any of the implementations described herein. In some implementations, a balance line **228** is fluidically arranged to transfer pressure from a portion of the wellbore downhole of the valve to the safety valve **112** to the safety valve actuation systems. Pressure from the balance line reduces a force needed to actuate the safety valve **112**. Alternatively or in addition, a pressurized canister **230** is included downhole near the safety valve **112**. The pressurized canister **230** provides pressure to the safety valve such that the pressure from the pressurized canister **230** reduces a force needed to actuate the safety valve **112**. Multiple types of compressed gas can be used, for example, compressed air, hydrocarbons, nitro-

gen, or carbon dioxide can be used. Other compressed gasses can be used without departing from this disclosure.

While primarily described and illustrated as a flapper **210**, other valve configurations can be used without departing from this disclosure. For example, a poppet style safety valve or a sliding sleeve safety valve can be used with similar bias and control mechanisms without departing from this disclosure. In some implementations, a temporary lock-out is included with the safety valve **112**. Such a lockout is used to keep the valve open during installation to allow the well to flow during installation. Once installed, the lock-out can be sheared, dissolved, or otherwise removed for standard operations.

In some implementations, the downhole receptacle **118** includes a latch **220** configured to retain or secure the electric submersible pump **102** within the downhole receptacle **118**. For example, collets **222** within the latch **220** include profiles that are configured to engage and mate with corresponding profiles on the electric submersible pump **102**. Such a latch can be hydraulically, mechanically, or electrically actuated without departing from this disclosure. In some implementations, the latch is integrated and retrievable with the electric submersible pump **102**. In some implementations, the latch is integrated and retrievable with the packer **119**.

In some implementations, the downhole receptacle **118** is a polished bore receptacle. In such an implementation, the electric submersible pump **102** includes annular seals **224** that seat against an inner wall of the downhole receptacle **118**. In some implementations, such seals can be included within the downhole receptacle **118** and seat against the electric submersible pump **102**.

FIG. 3A is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement **300**. The electrical submersible pump and safety valve arrangement **300** shares similar components and arrangements to the artificial lift arrangement **100b**, and, in some instances, can be used within the artificial lift arrangement **100b**. The electrical submersible pump and safety valve arrangement **300** is substantially similar to the electric submersible pump and safety valve arrangement **200** with the exception of any differences described or illustrated herein.

In this implementation, the safety valve **112** is integrated into the downhole receptacle **118**. As such, the shroud **208** of the electric submersible pump **102** is stabbed directly into the downhole receptacle **118**. In some implementations, the electric submersible pump **102** includes a hydraulic line that operates the safety valve **112** once the electric submersible pump **102** is stabbed into the downhole receptacle **118**. In such implementations, a hydraulic connection is made by the stabbing process. In some implementations, such a hydraulic connection requires a clocking feature to align the electric submersible pump portion of the hydraulic connection and the receptacle portion of the hydraulic connection. In some implementations, the hydraulic connection can include an annular hydraulic connection that does not require a clocking mechanism.

FIGS. 3B-3D are side cross-sectional diagrams of example packer and receptacle arrangements that can be used with aspects of this disclosure. In some implementations, the packer **119** includes an isolation barrier **302**. During installation of the packer **119**, in some implementations, the packer **119** is run with the isolation barrier **302** in an open position. Once the packer is installed, the isolation barrier **302** is then closed either responsive to the packer being set or by an independent slickline run. In such

implementations, a mechanical shifting tool **304** is included at a downhole end of the safety valve **112** and/or the electric submersible pump **102** (whichever is more downhole upon the package). The mechanical shifting tool **304** is configured to actuate the isolation barrier. When the safety valve **112** or electric submersible pump **102** are stung into the packer **119**, the isolation valve is opened, for example, by the mechanical shifting tool **304**, until the safety valve **112** and/or electric submersible pump **102** are removed from the packer **119**. The mechanical shifting tool **304** is shaped such as to allow fluid flow around or through the mechanical shifting tool **304**. For example, in the illustrated implementations, the mechanical shifting tool **304** includes a central body **306** with centralizers **308**. Uphole of the central body **306** and centralizers **308** is a perforated section **310** that allows fluid flow into the flow passage defined by the safety valve **112**. Such an arrangement allows for actuation of the isolation barrier **302** while maintaining production fluid flow. Other arrangements are feasible without departing from this disclosure. For example, in some implementations, the mechanical shifting tool **304** defines an inlet and flow passage. While primarily illustrated and described with relation to the safety valve arrangement **300**, the details illustrated and described in relation to FIGS. 3B-3D can be applied to any of the arrangements described within this disclosure.

FIG. 4 is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement **400**. The electrical submersible pump and safety valve arrangement **400** shares similar components and arrangements to the artificial lift arrangement **100b**, and, in some instances, can be used within the artificial lift arrangement **100b**. The electrical submersible pump and safety valve arrangement **400** is substantially similar to the electric submersible pump and safety valve arrangement **200** with the exception of any differences described or illustrated herein.

In the example electrical submersible pump and safety valve arrangement **400**, the packer **119** includes a nipple profile **402**. In operation, the safety valve **112** locks into the nipple profile **402** with a wetmate connection. That is, hydraulic and electrical connections are made downhole during installation. In implementations where the electric submersible pump **102** is not integrated with the safety valve **112**, the electric submersible pump **102** then snaps into similar wetmate connection on the body of the safety valve **112**. In some implementations, the wetmate connection includes circumferential seals and sealbores that allow communication to the safety valve **112**. In some implementations, the wetmate connections include a male protrusion stabbing into a female recess where there is sealing elements on one or both and the position of the male or female could be on the safety valve **112** or the pump **102**.

Also include the idea of the packer with the safety valve integral to the packer, such that there is a wetmate that the pump will connect with hydraulically or electrically. The pump snaps into the wetmate body in order to hold it in place.

FIG. 5 is a side cross-sectional view of an example electrical submersible pump and safety valve arrangement **500**. The electrical submersible pump and safety valve arrangement **500** shares similar components and arrangements to the artificial lift arrangement **100b**, and, in some instances, can be used within the artificial lift arrangement **100b**. The electrical submersible pump and safety valve arrangement **500** is substantially similar to the electric submersible pump and safety valve arrangement **400** with

the exception of any differences described or illustrated herein. The electrical submersible pump and safety valve arrangement **500** includes an inverted electric submersible pump **502**. That is, the electric motor **202** is uphole of the fluid end **204**. Such an arrangement is able to eliminate the shroud **208** (seen in FIG. 2) used in other implementations.

In an example operation, an electric submersible pump is received by a packer that includes a receptacle configured to receive the electric submersible pump. A safety valve is also received by the packer. In some implementations, the safety valve is received separately from the electric submersible pump. That is, the safety valve is integrated with the packer. In some implementations, the safety valve is received with the electric submersible pump. That is, the safety valve is integrated with the electric submersible pump.

In some implementations, the packer includes a latch. The electric submersible pump is secured to the packer by the latch. In some instances when the electric submersible pump is removed from the wellbore, the following steps are taken. An cable containing electrical and/or hydraulic lines is sheared at an uphole end of the electric submersible pump. The cable is sheared by a shearable interconnect at the electric submersible pump, and the shearable interconnect is triggered by an over-pull of the cable from the wellhead or topside facility. Once the cable is sheared, it is recovered from the wellbore. Once the cable is recovered, a fishing tool is received by the electric submersible pump. The fishing tool pulls up the electric submersible pump. In some implementations, the fishing tool performs an over-pull to release the electric submersible tool from the latch of the receptacle. In some implementations, a jarring tool is used to release the electric submersible tool from the latch of the receptacle. In some implementations, the latch has already been released prior to removing the electric submersible pump, for example, in implementations where the latch is hydraulically powered, the latch is released when the cable is sheared.

In implementations where the safety valve is integrated with the electric submersible pump, the fishing tool pulls the electric submersible pump and the safety valve out as a single unit. In implementations where the safety valve is integrated with the packer, the safety valve remains downhole with the packer once the electric submersible pump is removed.

While this disclosure has primarily describe electric submersible pumps, other downhole artificial lift systems, such as electric submersible compressors, top-driven pumps or compressors, plunger pumps and compressors, and gerotor pumps can be used without departing from this disclosure. Other mechanical lift devices, including positive displacement and centrifugal fluid movers, can be used without departing from this disclosure.

A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of this disclosure. For example, the safety valve can be integrated with an uphole end of an electric submersible pump without departing from this disclosure. Accordingly, other implementations are within the scope of the following claims.

What is claimed is:

1. An electric submersible artificial lift system comprising:
 - an electric motor;
 - a fluid end coupled to and configured to be driven by the electric motor;

- a shear interconnect at an uphole end of the electric submersible artificial lift system, the shear interconnect configured to shear a cable extending between the electric submersible artificial lift system and a topside facility, the shear interconnect configured to shear the cable at the electric submersible artificial lift system;
 - a safety valve disposed at a downhole end of the electrical submersible artificial lift system, the safety valve switchable between an open position and a closed position, the safety valve arranged to, while in the closed position, cease flow through the electric submersible artificial lift system and through a wellbore in which the electric submersible artificial lift system is installed; and
 - a downhole receptacle, wherein the electric submersible artificial lift system is configured to stab into and seal with the downhole receptacle disposed within the wellbore, wherein the safety valve is configured to switch position in response to the electric submersible artificial lift system stabbing into and sealing with the downhole receptacle.
2. The electric submersible artificial lift system of claim 1, further comprising a shroud encapsulating the electric motor and the fluid end.
 3. The electric submersible artificial lift system of claim 1, wherein the safety valve comprises a flapper valve biased towards a closed position.
 4. The electric submersible artificial lift system of claim 1, wherein the electric motor and the fluid end are coupled together by a magnetic coupling.
 5. The electric submersible artificial lift system of claim 1, wherein the safety valve is integrated into the electric submersible artificial lift system.
 6. The electric submersible artificial lift system of claim 1, wherein the electric motor is uphole of the fluid end.
 7. A method comprising:
 - receiving an electric submersible artificial lift system by a packer comprising a receptacle configured to receive the electric submersible artificial lift system;
 - receiving a safety valve by the packer; and
 - switching the safety valve from a closed position to an open position in response to receiving the safety valve by the packer.
 8. The method of claim 7, wherein the packer comprises a latch, the method further comprising:
 - securing, by the latch, the electric submersible artificial lift system to the packer.
 9. The method of claim 8, further comprising:
 - releasing, by the latch, the electric submersible artificial lift system.
 10. The method of claim 9, wherein releasing the electric submersible artificial lift system comprises:
 - receiving a fishing tool by the electric submersible artificial lift system.
 11. The method of claim 10, further comprising overpulling or jarring the electric submersible artificial lift system by the fishing tool, the latch releasing the electric submersible artificial lift system responsive to the over-pull or jar.
 12. The method of claim 11, further comprising leaving the safety valve within a wellbore with the packer.
 13. The method of claim 7, further comprising:
 - releasing a cable at an uphole end of the electric submersible artificial lift system.